

COMP5122M Data Science
Coursework 1

Global Climate Attitudes and Eco-Anxiety — Part I



Group Gloom

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Background and Aim

This report analyses whether there is a consistent age gap across countries in support for renewable energy transition using the dataset *United Nations Development Programme (UNDP) Peoples’ Climate Vote 2024* [4]. The data contains responses to various climate change related questions, representing over 73,000 people across 77 countries. Country-level breakdowns of the responses are available for 73 of the countries, with Kazakhstan, the Netherlands, Poland and Sweden included only in global estimates due to small country samples sizes.

Methodology and Preprocessing

Data were analysed in Python using `numpy`, `scipy` and `pandas`, with visualisations created in `matplotlib` and `seaborn` for reliable analysis.

The survey question: “How quickly should your country replace coal, oil, and gas with renewable energy, such as power from the wind or sun” was isolated as the primary focus, as it directly addresses the core research objective. The dataset was therefore filtered to retain only responses to this question. Pre-processing steps included removing aggregate columns (“Global” and “All ages”) and prior to sentiment scoring, weighted means were renormalised to account for excluded “Don’t know” responses ($\approx 20\%$ of total responses). For each response category i , the renormalised weighted mean WM_i^* is calculated as:

$$WM_i^* = \frac{WM_i}{100 - WM_{DK}} \times 100,$$

where WM_{DK} represents the weighted mean of response “Don’t know” and WM_i represents the weighted mean per valid response.¹

Cross-country comparison of the degree of urgency with which people prefer the transition to renewable energy was based on a *Sentiment Score* for each country c and age group a . The response categories were mapped to an equidistant scale, and the score was computed as:

$$\text{SentimentScore}_{c,a} = \sum_{i=1}^4 p_i S_i, \quad \text{where } S_i = \begin{cases} 1 & \text{“Very quickly”,} \\ \frac{2}{3} & \text{“Somewhat quickly”,} \\ \frac{1}{3} & \text{“Slowly”,} \\ 0 & \text{“Not at all”.} \end{cases}$$

Age gap comparisons focused on the 18–35 and 36–59 groups only, as no country-level data were available for respondents under 18, and data for older groups were too sparse for reliable analysis.

The “**Age Gap**” metric was calculated as the difference between the two groups’ scores. Positive values indicating stronger support among younger respondents, negative values indicating stronger support among middle-aged respondents, and values near zero indicating little age difference.

Exploratory Analysis Results

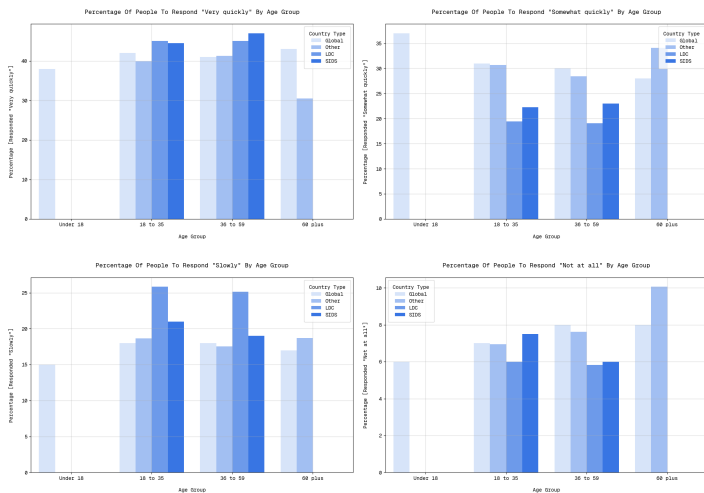


Figure 1: Graphs showing the percentage of people who responded decisively across the four categories of country type, grouped by age.

Nine of the sampled countries were classified as Small Island Developing States (SIDS) and nineteen were classified as Least Developed Countries (LDCs). The “Global” category serves as an aggregate baseline representing worldwide averages, allowing for comparison between focused country types and global trends. Grouped bar charts facilitate direct comparison of response distributions across demographic groups.

Figure 1 shows a noticeable pattern: where an emerging trend of stronger support for rapid transition were demonstrated by LCDs and SIDS compared to developing nations, implying economic development status may influence climate-change appetite.

¹Percentages in the dataset are rounded to the nearest whole number, therefore totals sometimes equal 99% instead of 100%. For relational consistency across categories, totals are assumed to be 100% when applying the renormalisation formula.

Cross Country Comparison of the Widest and Narrowest Age Gaps

Heatmaps were chosen to enable rapid identification of cross-country patterns, while caterpillar plots illustrate distribution characteristics. Figure 2 was used to identify key outliers in the dataset; the largest are as follows:

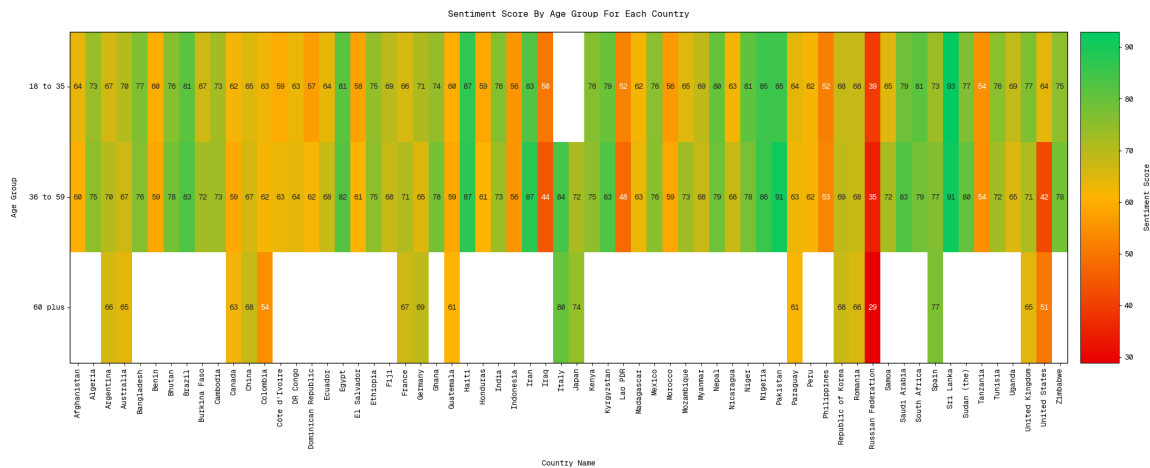


Figure 2: Heatmap showing the relationship between country sentiment scores amongst age groups 18 to 35, 36 to 59, and 60 plus.

- **The Unites States of America** acts as a significant, negative outlier demonstrating a fierce disparity in opinion towards climate change between younger and older demographics within the country with a range of 22% between the most (18 to 35) and least (36 to 59) supportive age categories.
- **The Russian Federation** presents as an extreme, negative outlier across all age groups when compared globally. The results implied that the population across the tested age categories harbours a particularly low appetite for climate-action, with the lowest reading of all countries at 29%.
- **Sri Lanka** upholds a significant, positive outlier; high sentiment scores were shown for both age demographics polled, with the highest globally at 93%. The results imply that the majority of the represented population garner strong climate change support.

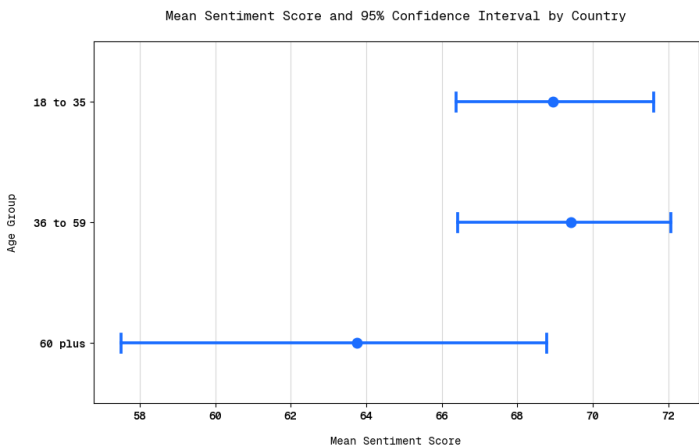


Figure 3: Caterpillar plot showing the global sentiment score confidence interval across age groups 18 to 35, 36 to 59, and 60 plus.

Figure 3 spotlights global trends across the age groups. The results convey that the oldest age group (60 plus) had the lowest mean sentiment score at ≈ 64 , implying the age group believed that climate change was of lower urgency than younger groups.

Furthermore, the “60 plus” group exhibited the widest spread of results, with the 95% interval spreading from ≈ 58 to 69. This range was far larger than either of the younger age groups, implying that the sentiment towards the renewable energy transition was far more varied amongst older demographics. However, due to missing country data for this demographic, analysis of this spread was left limited and incomplete, as a small sample size cannot be wholly representative.

Consistency of Age Gaps in Support for Renewable Energy Transition

Figure 4 further helps visualise country-level outliers in differences between age demographics in sentiment towards climate change¹. The United States shows the largest age-based disparity, with younger respondents expressing far stronger support for the renewable energy transition than older groups.

SIDS countries were also investigated, showing significant, negative sentiment disparity between the older and younger demographics. The data conveys the opposite message compared to the global trend, in that, the older demographic within the island nations view climate change as *more* urgent.

¹“Age Gap” on Figure 4 refers to the sentiment–difference metric between age groups, as defined in the methodology on page 1.

Age Gap For Each Country

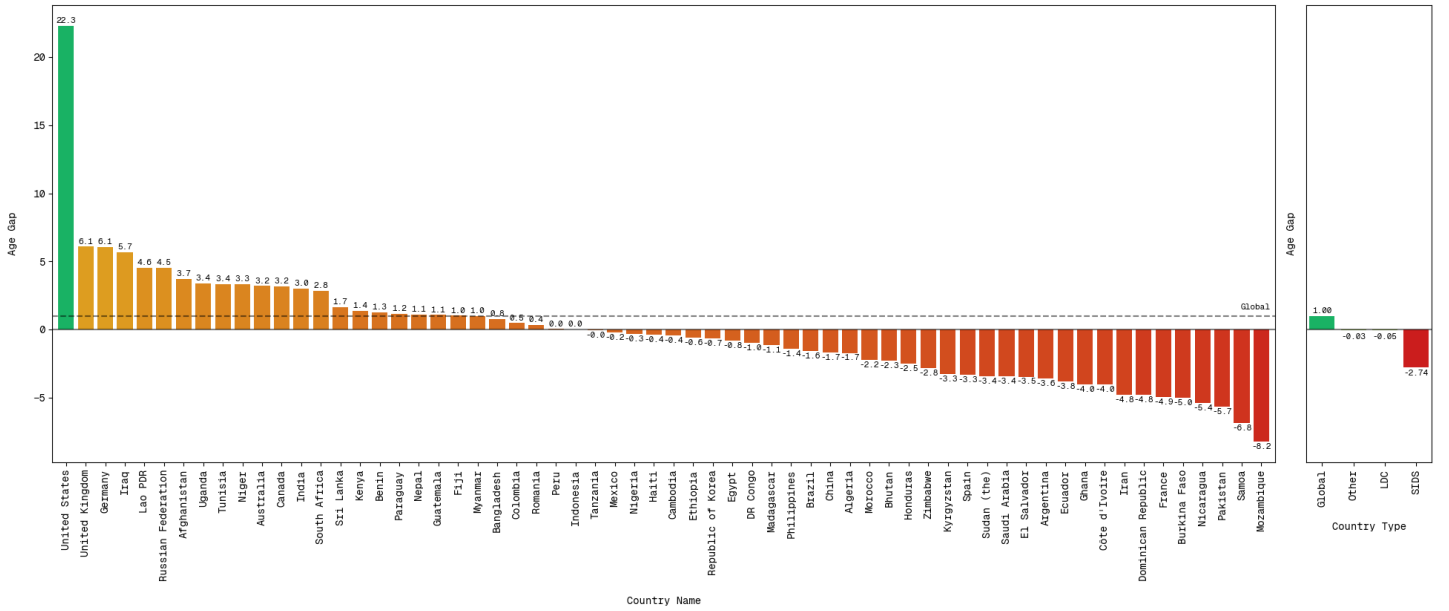


Figure 4: Graph showing the relationship of the “Age Gap” metric to each country.

The overall distribution of “Age Gap” values reflects this mixed landscape: the mean (-0.21) and median (-0.38) lie close to zero, indicating no consistent global direction, but the wide spread (standard deviation 4.37, ranging from -8.20 to 22.31) shows substantial variation between countries.

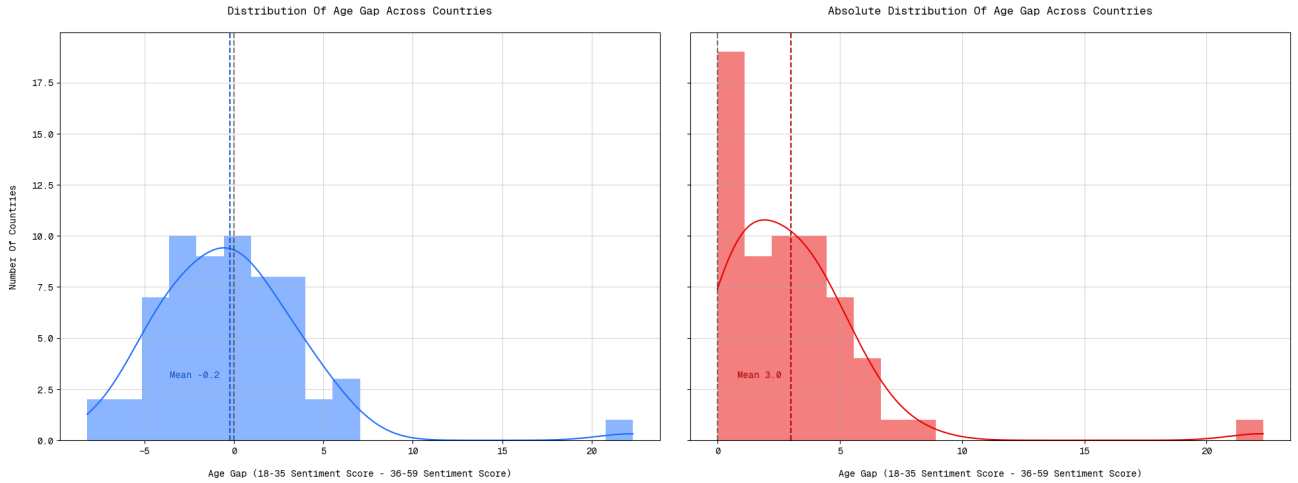


Figure 5: Graph showing the distribution of the Age Gap metric across countries.

By taking the absolute value of the “Age Gap” in Figure 5, the difference between the demographics is observed to coalesce around a mean of ≈ 3.0 . To quantify the overall magnitude of generational differences, a one-sample t-test was conducted on the absolute “Age Gap” values across all countries, ignoring the direction of the difference to evaluate whether the size of the “Age Gap” differs significantly from zero, irrespective of whether younger or older respondents show stronger support. The test produced a t-statistic of 7.483 and a p-value of < 0.001 , indicating strong evidence that the average absolute “Age Gap” is significantly greater than zero suggesting that while the global average (signed) “Age Gap” may not show a consistent direction, sentiment towards climate change depends significantly on generation across countries.

Conclusions

The global pattern suggests that while younger generations generally express stronger urgency for renewable transition, the magnitude of this gap varies significantly by national context, influenced by economic development, media landscapes, and political polarisation. These findings are consistent with previous research suggesting a generational divide on attitudes to climate change in the United States of America [1]. In contrast, in countries where climate impacts are more immediately felt, such as LDCs and SIDS, generational differences may reflect direct lived experience instead[3]. Interpretation should also consider potential limitations arising from data scarcity in certain countries and from converting categorical Likert-scale responses into numerical scores, as the meaning or strength of these categories may differ across cultural or regional contexts[2].

Appendix

- [1] Cary Funk. Key findings: How americans' attitudes about climate change differ by generation, party and other factors. *JOURNAL*, 42(12):1150–1152, 2021.
- [2] Jerry W. Lee, Patricia S. Jones, Yoshimitsu Mineyama, and Xinwei Esther Zhang. Cultural differences in responses to a likert scale. *Research in Nursing & Health*, 25(4):295–306, 2002. Accessed: 2025-11-14.
- [3] Landlocked Developing Countries Office of the High Representative for the Least Developed Countries and Small Island Developing States (UN–OHRLLS). The impact of climate change on the development prospects of the least developed countries and small island developing states. Technical report, United Nations, 2009. Accessed: 2025-11-14.
- [4] United Nations Development Programme and University of Oxford. Peoples' climate vote 2024 data center. <https://peoplesclimate.vote/data-center>, 2024. Accessed: 2025-11-14.